MULTI-VENDOR ONLINE MARKETPLACE WITH ORDER WORKFLOW ENGINE

1. Abstract

The Multi-Vendor Online Marketplace with Order Workflow Engine is a web application developed to bring different sellers and customers together on a single platform. The system allows sellers to manage their products and stock, while customers can purchase products from different sellers through one shopping interface. The main focus of the project is to manage the complete order process, from checkout to delivery. When an order contains products from multiple vendors, the system keeps track of each item and allows the respective vendors to process their orders independently. It also maintains stock details, records order updates, and provides different access permissions for administrators, vendors, and customers. The application is developed using HTML, css, js, Python, Django REST Framework, and MySQL. The project aims to make multi-vendor shopping easier to manage by keeping product information, inventory, and order activities organized in one place.

2. Objective

The objective of this project is to develop an online marketplace that supports multiple sellers and manages their order activities through a common system. It aims to provide a simple way for vendors to maintain products and stock while allowing customers to purchase items from different sellers. Another objective is to manage order progress properly so that each vendor can handle their assigned products without affecting the processing of other vendors' items. The system also aims to maintain accurate stock records, control user access, and provide clear information about order status.

3. Existing System

Many online shopping applications support product listings, payments, and order tracking. However, when a marketplace involves multiple independent sellers, the system must coordinate their separate order-processing activities. For example, products in the same customer order may be handled by different vendors and may reach the customer at different times. Inventory changes, cancellations, and delayed status updates also need to be managed carefully. Without suitable workflow rules, the application may show confusing order progress or maintain incorrect stock records.

Drawbacks and Challenges

- Products belonging to different vendors may require separate processing and delivery updates.
- A single overall order status may not clearly represent the progress of every product.

- Simultaneous purchases can cause stock inconsistencies if inventory updates are not handled safely.
- Vendors need access to their own orders without being able to modify other vendors' records.
- Order cancellations must be handled carefully to avoid incorrect stock restoration.
- Administrators need a common view of vendor activities and order progress.

4. Proposed System

The proposed system provides a common shopping platform with separate workspaces for administrators, vendors, and customers. Vendors can submit registration details for approval and manage their products after approval. Customers can browse products, add items from different sellers to the cart, and place orders through a common checkout process.

The main feature is the Order Workflow Engine, which manages the progress of individual items within an order. It records order creation, vendor processing, shipping, delivery, and permitted cancellations. The system also checks product availability during checkout and uses controlled database operations to maintain accurate stock quantities. Each vendor can update only the order items assigned to them, while customers can view the progress of their purchases. The administrator can monitor marketplace records and manage vendor access.

5. Users & Roles and Responsibilities

Customer

- Create an account and log in.
- Search and compare products from different vendors.
- Add products to the cart and place orders.
- Make payments and view payment status.
- Track individual products within an order.
- Request cancellation when permitted.
- Replacement of the product if required.
- Analytical Dashboard for the customer expenditure.

Vendor

- Register and submit details for approval.
- Add, update, and deactivate products.
- Maintain available stock quantities.
- View orders containing their products.
- Process assigned items and update their status.
- Add new vendors to the group.
- Analytical Dashboard for the product sells and revenue.

Administrator

- Review vendor registrations and approve or reject them.
- Manage customer and vendor accounts.
- Maintain product categories and monitor product listings.
- View marketplace orders and their progress.
- Review inventory information and basic activity reports.
- Dashboard for both the customer and vendors.

7. Hardware Requirements

- Processor: Intel Core i3 or equivalent.
- RAM: Minimum 4 GB; 8 GB recommended.
- Storage: At least 10 GB of free disk space.
- Input Devices: Keyboard and mouse.

8. Software Requirements

- Operating System: Windows or Linux.
- Frontend: React JS, HTML, CSS, and JavaScript.
- Backend: Python and Django.
- API Development: Django REST Framework.
- Database: MySQL.

- Code Editor: Visual Studio Code.
- Browser: Google Chrome, Microsoft Edge, or another modern browser.
- Version Control: Git and GitHub.